LIN7076 – Foundations of Computational Linguistics

Formal languages

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Deep intuitions about well-formedness

(1) a. Colorless green ideas sleep furiously.

b. * Furiously colorless green sleep ideas.

'Twas brillig, and the slithy toves
Did gyre and gimble in the wabe:
All mimsy were the borogoves,
And the mome raths outgrabe.
"Beware the Jabberwock, my son!
The jaws that bite, the claws that catch!
Beware the Jubjub bird, and shun
The frumious Bandersnatch!"

Jabberwocky, Lewis Carroll

Plan for today

Can we come up with simple rules summarizing a language or a fragment thereof?

- We'll take a break from statistics and algebra to talk a bit about **formalizing language structure**.
- We'll explore two main approaches: **regular expressions**, **context-free grammars**.
- We'll discuss how these can be used for concrete applications: find-and-replace, parsing and structure-mining.
- We'll also review some aspects of natural language that are fundamentally challenging for these approaches.

Examples of ungrammaticality/ill-formedness

- Tensed embedded clause cannot be subject-less ("ban on hyperraising"):

(2) a. * Jo seems [that won]. b. Jo seems [to have won].

- Subjects and main verbs must agree:

(3) a. * The dogs barks. b. The dog barks.

- Adverbs must precede main verbs:

(4) a. * Jo eats often apples. b. Jo often eats apples.

- All these generalizations are independent of meaning–they are **syntactic**, and sometimes language-dependent.
- Violating them causes ungrammaticality/**ill-formedness**.

Examples of infelicity

- A proposition and its negation cannot be conjoined (contradictory!) or disjoined (tautology!):

(5) a. # Jo is inside and outside.

b. # Jo is inside or outside.

- Sentences should take obvious facts for granted...

(6) a. # A sun is shining.
⇝ There's at least one sun.

b. The sun is shining.
⇝ There's a unique sun.

- But should not take new information for granted:

(7) a. ?? Jo will need to bring **her alligato**r to the vet.

b. Jo **has an alligator**, and will need to bring it to the vet.

- These generalizations depend on meaning (**semantic**) and largely language-independent.Violating them causes **infelicity**.

Formalizing well-formedness

- Today we'll **focus on modeling grammaticality/well-formedness**.
- A language can be modeled as **the set of all the sentences that are well-formed in that language–independently of meaning**.
- For instance, English = {The dog barks, Colorless green ideas sleep furiously, ...}. How big is this set?

Other possible specifications of a “language”

- More abstractly, a language can be seen as the set of well-formed sequences of syntactic categories (instead of words instantiating these categories).
- In that case, English = {Det Noun Verb, Adj Adj Noun Verb Adv, ...}. How big is this set?
- More narrowly, a language can be seen as the set of well-formed sequences of phonological elements (e.g. consonants C, and vowels V[1])
- English is hard to illustrate, so let’s take Japanese whose building blocks are mostly CV syllables plus occasional nasals. In that case Japanese = {CV, CVCV, CVCVCV, CVnCV, CVCVn, ...}.
- In all these cases, we are interested in **well-formed units of a language** (strings of words; strings of syntactic labels; strings of phonemes...).

[1] One can choose more specific categories too!

The need for recursivity

- To model natural languages, formal languages need to generate **an infinite number of outputs from finite means**.
- Finite means:
  - A **finite alphabet** Σ of primitive "words" (lexical items, or syntactic categories).
  - A **finite set of rules** with some degree of **recursivity**.
- Recursivity is the **capacity for a rule to refer to itself**. Because recursive rules may apply an arbitrary number of times, an infinity of outputs can be generated.
- Concrete example: a folder-based file system is inherently recursive. Any **folder** may contain only **files** ("base case"), or, at least one other **folder** ("recursive case").
- More generally, anything that displays an **arborescent structure** will be recursive!

A toy example: intensified adjectival phrases

- Suppose our alphabet Σ is { a, the, very, tall, girl, boy }. What kind of well-formed phrases can we build out of this alphabet?
- *the boy* and *the girl* are well-formed.
- *a boy* and *a girl* are well-formed.
- *the/a tall boy/girl* are well-formed.
- *the/a very tall boy/girl* are well-formed.
- *the/a very very tall boy/girl* are well-formed.
- *the/a very very very tall boy/girl* are well-formed.
- ...
- Where do we spot recursivity here?

Regular expressions

- Let Σ be an alphabet. We call the members of Σ **words** (they can be "real" words like *cat*, syntactic labels like Det, phonological units like V...).
- Regular expressions (**regexes**) are used to specify **word patterns**.
- Regexes are built out of consist of **constants**, which denote sets of words, and **operator symbols**, which denote operations over these sets.
- Constants: $\emptyset$ (the empty regex), and $\{\epsilon\}$ (the empty-word regex) are regexes. For any w in Σ, $\{w\}$ is a regex.
- Operations: given R and S two regexes...
  - RS is the pairwise **concatenation** of R and S and is a regex.
  - $R|S$ is the **union** of R and S and is a regex.
  - R^* is the set containing ϵ and all possible concatenations of elements of R ("**Kleene star**") and is a regex.
- We'll use parentheses to signal the order of precedence between these operations.

A regex for intensified adjectival phrases

- Σ = { a, the, very, tall, girl, boy }.
- We want a regex to match the set of expressions {a boy, a girl, a tall boy, a tall girl, a very tall boy, a very tall girl, a very very tall boy, ... the boy, the girl, ...}.
- The phrase starts with *the* or with *a*. Achieved with the **union** operation (a|the).
- The phrase continues with an arbitrary number of *very* (including zero) followed by *tall*; or no adjective at all. The former option is modeled by **concatenating** the **Kleene star** of *very* (very*) with *tall* to form (very*)tall. The latter option is simply ϵ (the empty word). We then union these two options ((very*)tall|ϵ).
- The phrase ends with *boy* or *girl*. Achieved again with the **union** operation (boy|girl).
- **Concatenating** all these parts together form the desired pattern: (a|the)((very*)tall|ϵ)(boy|girl).
- What is the issue with the regex (a|the)(very*)(tall|ϵ)(boy|girl)?

Useful shorthands

- One can define useful **shorthands** from the three core operations (concatenation, union, Kleene star). If R and S are regexes:
  - R^+ is defined as RR^*: it is the set all possible concatenations of elements of R.
  - R^k with $k > 0$ a finite integer is defined as $\underbrace{RRR...R}_{k \text{ times}}$ is the set of all possible concatenations of elements of R involving exactly k elements.
  - $\mathrm{R}^{\{k,l\}}$ with $l > k > 0$ finite integers is defined as $R^k|R^{k+1}|R^{k+2}|...|R^l$ is the set all possible concatenations of elements of R involving between k and l elements.
  - . (wildcard) stands for any element of the primitive alphabet Σ.
  - R? is defined as $(R|\epsilon)$ and is the set of elements of R plus the empty word.

Practice: right-embedded complementation

- Suppose $\Sigma = \{\text{NP, V, VP, that}\}$.
- We want to model sentences of the form *Jo believes that Al thinks … that Lu imagines that Ed is happy*, where *Jo*, *Al*, *Lu*, …, *Ed* are NPs, *believes*, *thinks*, …, *imagines* are V, and *is happy* is a VP.
- What's the pattern of *Ed is happy*? NP VP.
- Of *Lu imagines that Ed is happy*? NP V that NP VP.
- Of *Al thinks that Lu imagines that Ed is happy*? NP V that NP V that NP VP.
- Of *Jo believes that Al thinks that Lu imagines that Ed is happy*? NP V that NP V that NP V that NP VP.
- The general pattern is: NP V that 0 or more times, followed by NP VP.
- Regex: (NP V that)*(NP VP)

Applications

- Mining a dataset for specific verbs and all their **possible inflections** (including noisy/erroneous ones!), for instance CHILDES when studying the acquisition of verbal predicates.
- Mining a naturalistic corpus for a wide range of **contracted/colloquial forms**; for instance when studying language change or sociolingusitic variation.
- Looking for specific **coocurrences separated by some arbitrary material**, e.g. perception verbs followed at some point by gerunds (*I see the girl with an umbrella dancing*); adjectival predicates followed by infinitives...
- Elaborate **PoS-tagging/parsing** based on morphological cues (e.g. suffixes *-ation*/*-ational*/*-ationalization*...).
- Automatic and conditional "**find-and-replace**" based on a pattern rather than one single string. See the Unix regex-based commands `grep` (find) and `sed` (find-and-replace).

Graphical representation: automata

- Regexes can be represented using diagrams called **automata**.
- Automata have **states**, represented with **circles**. There's one **initial** state (where you start), and at least one **accepting** state (where you want to end, signaled with a double circle).
- States are connected with arrows labeled with words of the alphabet Σ (including the empty word ϵ).
- An automata **matches a sentence** if there is a path from the initial state to an accepting state whose successive arrows match the words of the sentence.

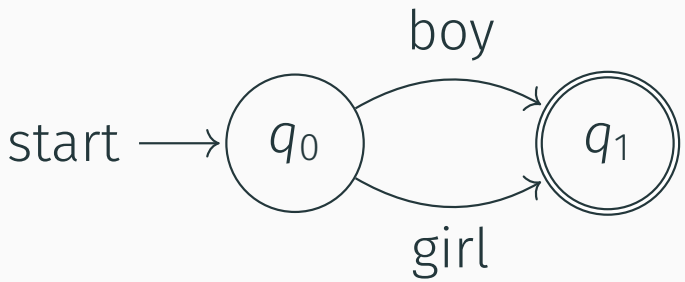
Concatenation automaton

- The following automaton accepts only one sentence, *the boy*. It corresponds to the concatenated regex the boy.

- More generally, regex **concatenation** is represented by **arrows between states**.

Union automaton

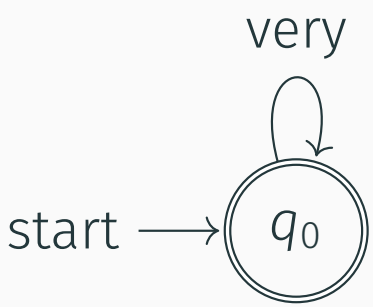
- The following automaton accepts *boy* or *girl*, i.e. it corresponds to the regex union operation (boy|girl).



- More generally, regex **union** is represented by **multiple arrows** originating from the same state.

Kleene star automaton

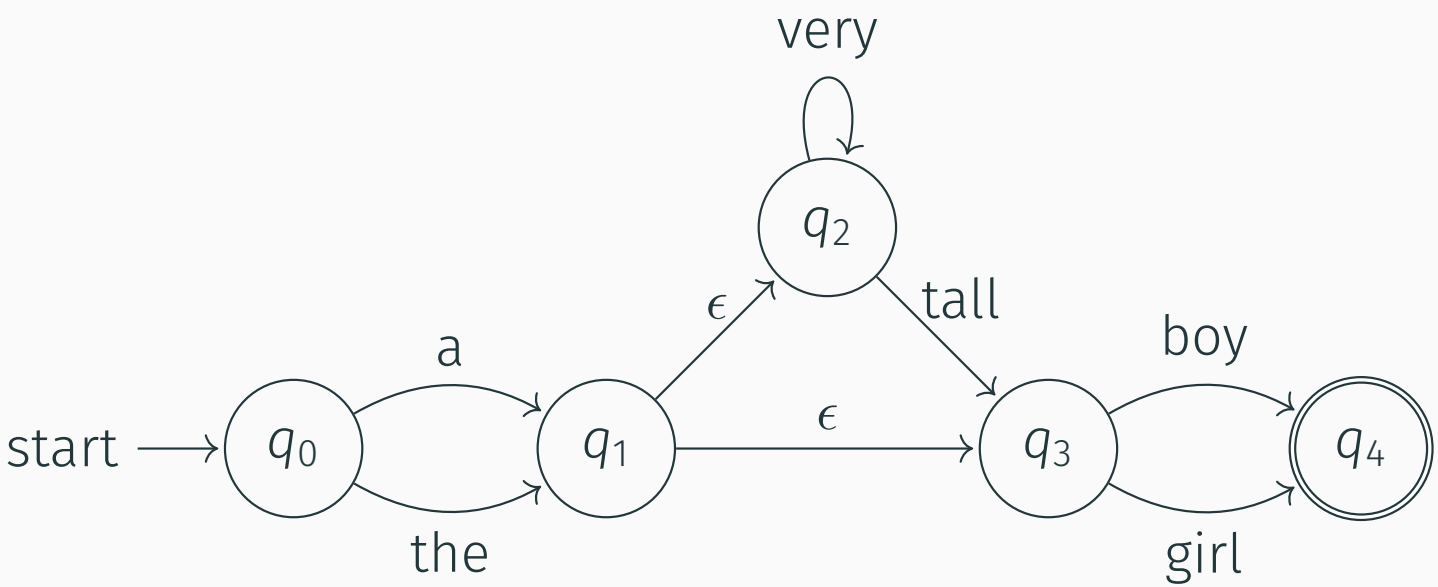
- The following automaton accepts ϵ, *very*, *very very*, *very very very*...i.e. it corresponds to (*very**).



- More generally, **Kleene star** operations are **feedback loops** to "previous" states.

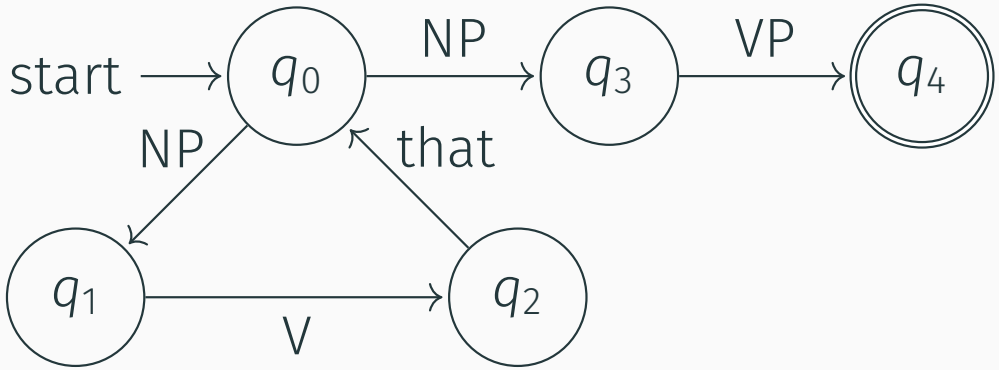
Practice: automaton for intensified adjectival phrases

- Recall that our pattern for *the/a boy/girl*, *the/a tall boy/girl*, *the/a very tall boy/girl*... was: (a|the)((very*)tall|ϵ)(boy|girl).
- It meant that first we want a determiner (*a* or *the*), then we want either an optionally intensified adjective (*very very ... very tall*) or nothing, and lastly, we want a noun (*boy* or *girl*).
- What's the automaton for this?



Practice: automaton for right-embedded complementation

- Recall that our pattern for *Jo believes that Al thinks… that Ed is happy…* was: (NP V that)*(NP VP).
- It meant that first we want NP V that (an embedding sentence) zero or more times, and then we finish by NP VP (the last embedded sentence).
- What's the automaton for this?



Linguistic challenges for regular expressions and automata

Structural ambiguities

- Compare:

(8) a. Tall boys and girls.
b. Tall boys and tall girls.
c. Tall boys and short girls.
d. Boys and tall girls.

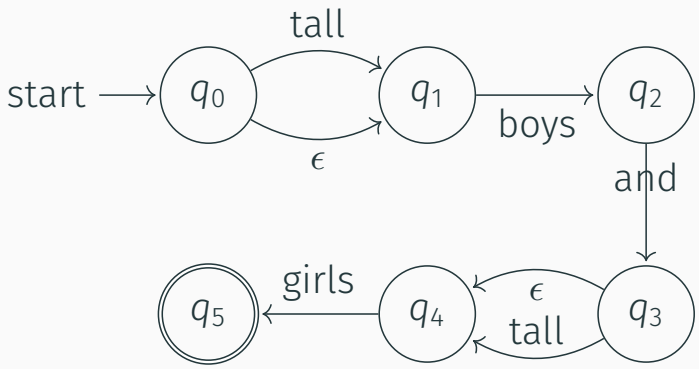
- When an adjective precedes conjoined NPs, as in (8a), **it's ambiguous whether the adjective applies to both NPs** (as in (8b)), **or to the first NP only** (with a meaning compatible with (8c)).
- This ambiguity disappears if the adjective follows the first and precedes the second NP, as in (8d).
- Similar-ish ambiguities arise in sentences like (9) and (10).

(9) The photographer saw the bird on the hill.

(10) They are building blocks.

Regexes/automata have a “weak” generative capacity

- Regexes/automata *can* match strings like *tall boys and girls*, *tall boys and tall girls* etc: (tall|ϵ) boys and (tall|ϵ) girls.



- But nothing in the traversal of the automata, or in the denotation of the regex (a sets of strings!) tells us that *tall boys and girls* can be understood either as *[tall [boys]] and [girls]* or as *[tall [boys and girls]]*!
- These devices capture the well-formedness of the string, but **nothing about its internal structure** (which influences meaning).
- In that sense, regexes and automata have a **weak generative capacity** w.r.t. structural ambiguities.

Deeper challenge: center-embedding

- Let's consider nested subject-modifying, object-gap relative clauses:

(11) a. The mouse is scared.
b. The mouse [that the cat chases] is scared.
c. The mouse [that the cat [that the dog barks at] chases] is scared.
d. The mouse [that the cat [that the dog [that the postman avoids] barks at] chases] is scared.

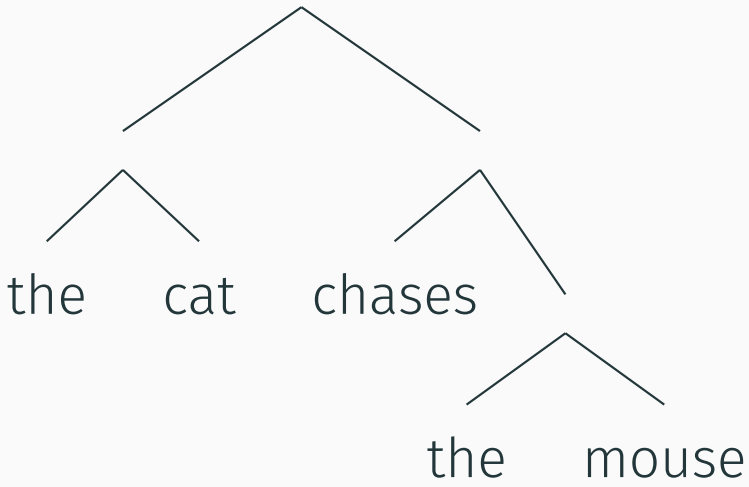
- The corresponding structures are: NP VP; NP that NP V VP; NP that NP that NP V V VP; NP that NP that NP that NP V V V VP... etc.
- Can we synthesize this using one single regex?
- The pattern appears to be: NP (that NP)^k (V)^k VP, **for any** $k \geq 0$. But there are an infinity of such k, so our "regex" would need to be infinitely long... Not a regex!

Challenges summary

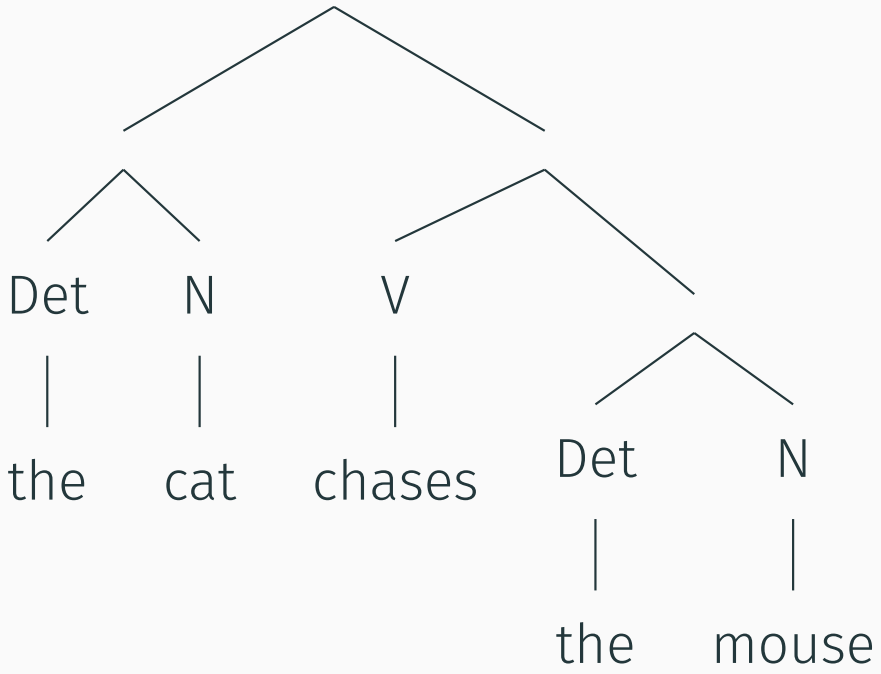
- The fundamental limitation of regexes/automata is that they **cannot keep track of arbitrary occurrence counts**. Anything that looks like a sequence of k identical patterns, followed by a sequence of k (or $k-1$, or $k+6$...) other patterns, cannot be matched by regexes/automata. They are **not expressive enough**.
- This sheds light on a **fundamental complexity difference between right-embedding** (*Jo thinks that Ed believes that...*), which *can* be captured by regexes, **and center-embedding** (*The mouse that the cat that... chases is scared*), which cannot.
- We have also seen that regexes and automata **cannot account for the internal structure of strings**, and the meaning differences that may arise from structural ambiguities.
- We now introduce context-free grammars to encode more structure and recursivity.

Context-free grammars

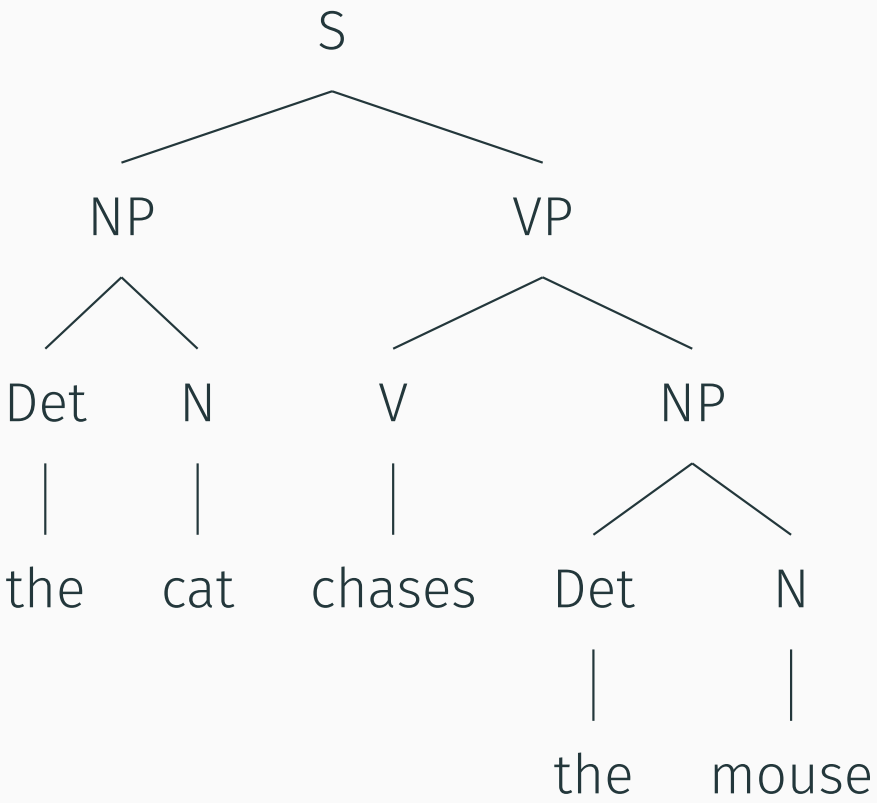
Syntax trees



Syntax trees



Syntax trees

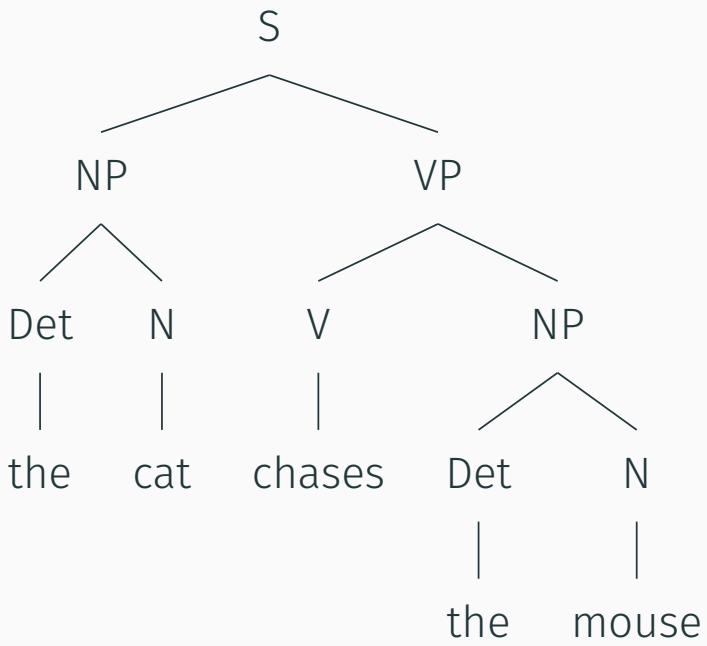


Generating syntax trees

- One can use context-free grammar to generate syntax trees. A context-free grammar is defined as:
  - A finite set Σ of **terminals**, (leaves, i.e. actual words of the language).
  - A finite set V of **nonterminals** (intermediate nodes, i.e. phrase or category labels). V contains a special nonterminal S used to represent entire sentences (root).
  - A set R of **production rules**, i.e. input $\rightarrow$ output pairs, where inputs are single nonterminals (understood as mother nodes) and outputs are (non)terminals or concatenations thereof (understood as children node(s)). For instance, 

corresponds to the production rule S $\rightarrow$ NP VP
- Pāṇini, a scholar in ancient India, came up with a grammar of Sanskrit following a very similar formalism in the mid-1st millennium BCE.

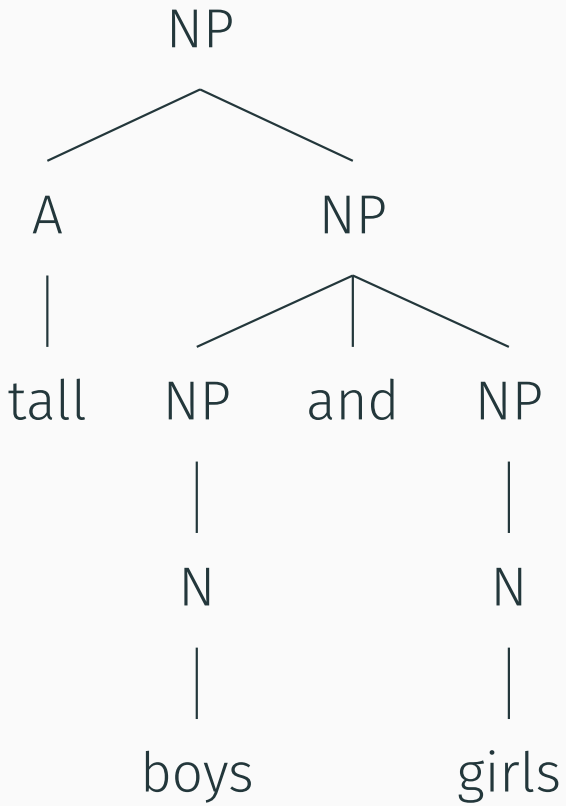
Production rules practice

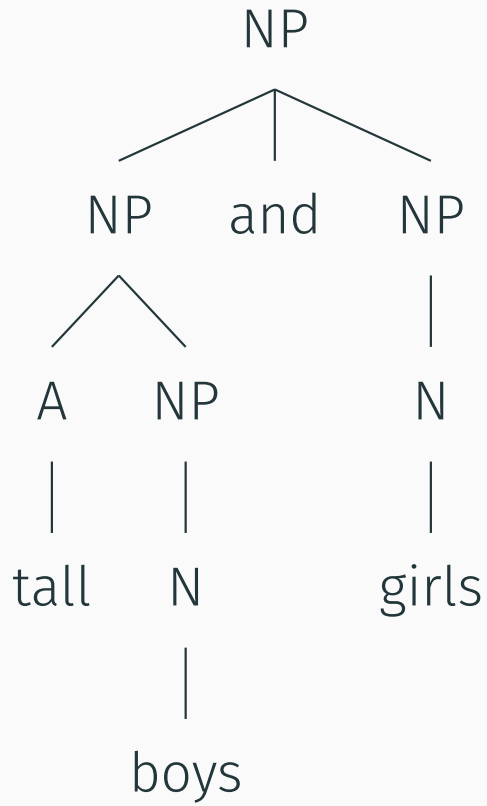


- What are the terminals? *the*, *cat*, *chases*, *mouse*.
- What are the nonterminals? S, NP, VP, Det, N, V.
- What are the production rules? S $\rightarrow$ NP VP; NP $\rightarrow$ Det N; VP $\rightarrow$ V NP; Det $\rightarrow$ the; N $\rightarrow$ cat; N $\rightarrow$ mouse; V $\rightarrow$ chases.
- What else can we generate from those rules? *The mouse chases the cat*.

Structural ambiguity 1: adjective attachment

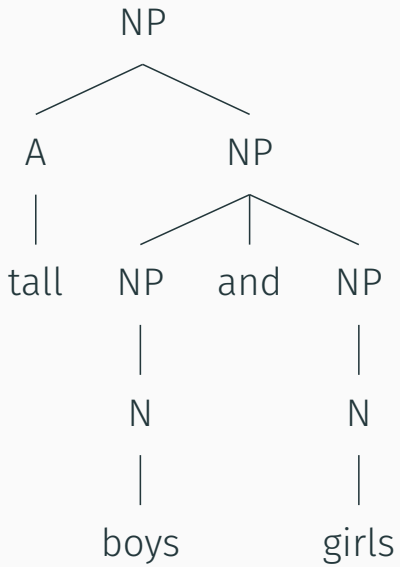
- Intuitively, which tree corresponds to which interpretation?

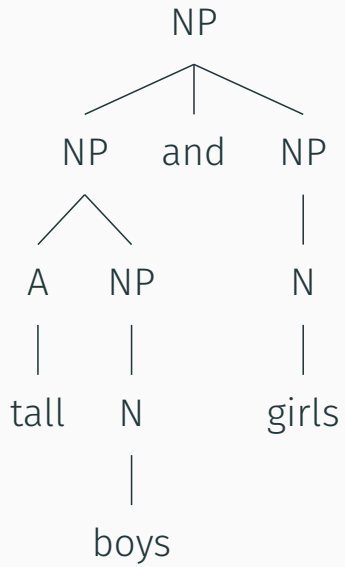




- Terminals? Nonterminals? Production rules?

Structural ambiguity 1: adjective attachment

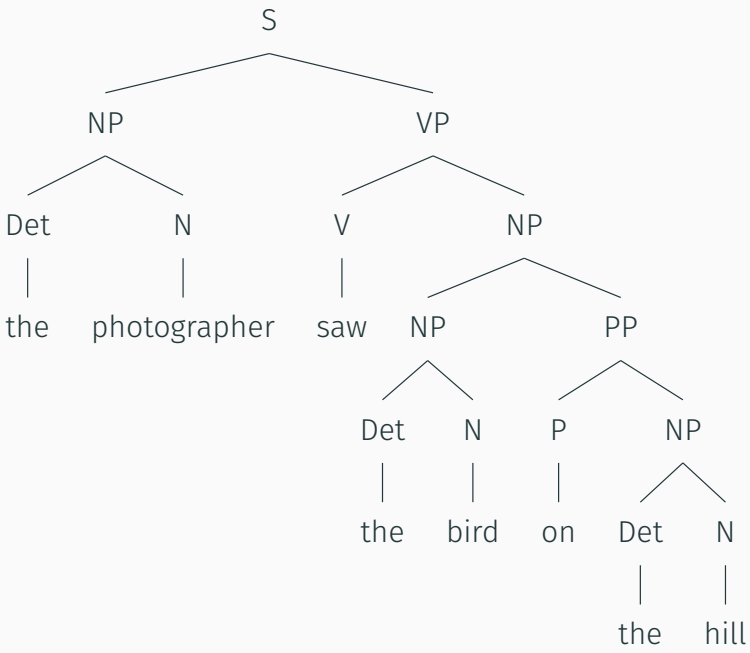


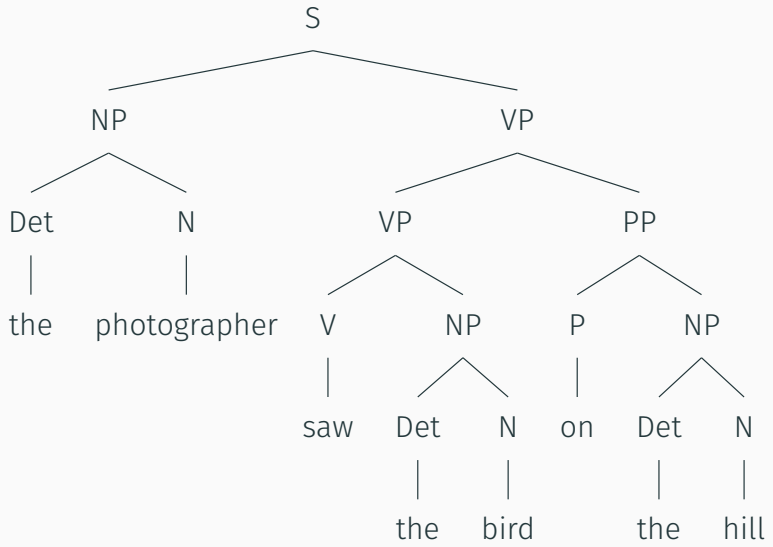


- Terminals? *tall*, *boys*, *and*, *girls*.
- Nonterminals? NP, A, N.
- Production rules? NP $\rightarrow$ A NP; NP $\rightarrow$ NP and NP; NP $\rightarrow$ N.
- What else can we generate? *Boys and tall girls*, *Tall girls and boys*, *Tall tall tall boys and girls*, *Tall boys and tall girls*, *Boys and boys and girls*... Slight **overgeneration issues**!

Structural ambiguity 2: PP-attachment

(9) The photographer saw the bird on the hill.





- Which tree intuitively yields which reading?

Structural ambiguity 2: PP-attachment

- Consider the grammar below. Which reading(s) does it capture?

S $\rightarrow$ NP VP
NP $\rightarrow$ Det N
VP $\rightarrow$ V NP
NP $\rightarrow$ NP PP
PP $\rightarrow$ P NP

Det $\rightarrow$ the
N $\rightarrow$ photographer, bird, hill
V $\rightarrow$ saw
P $\rightarrow$ on

- The only rule in which PP is an output (daughter node) is NP $\rightarrow$ NP PP. In that rule, PP's sister node is NP, which means PP attaches to an NP. It models the *bird on the hill*-reading.

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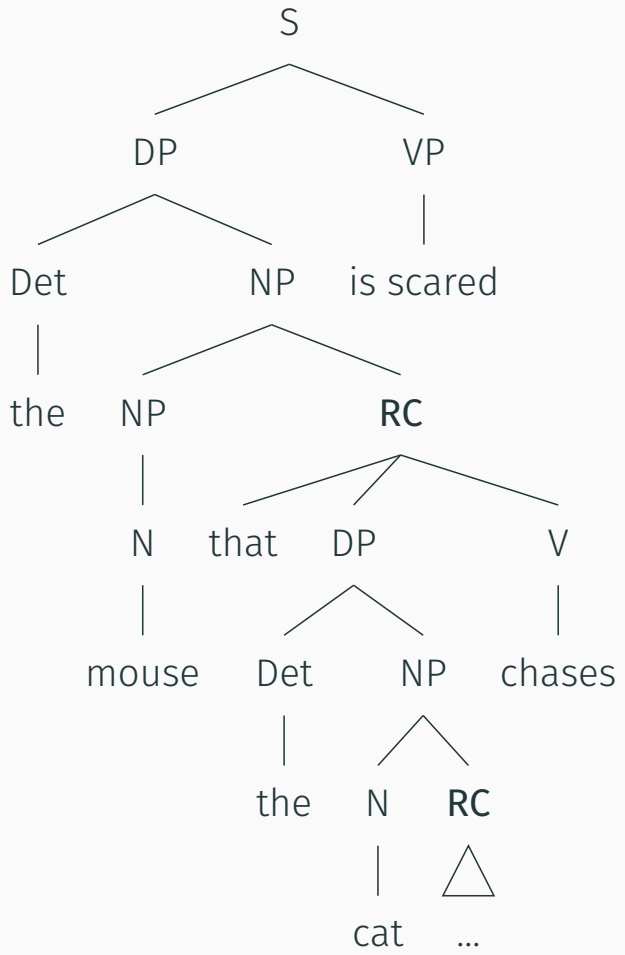
Context-free grammar and structural ambiguities recap

- Context-free grammars primarily generate strings by **calling production rules** (i.e. branching), until all the outputs are terminals.
- But interestingly, keeping track of which production rules were used to generate the final string (and what was their exact timing), sometimes allows us to **derive distinct syntax trees for the same string**–cashing out semantic ambiguities.
- In that sense, context-free grammars have a **strong generative capacity** when it comes to adjective and PP attachment ambiguities.
- We proceed to show that context-free grammar also capture center-embedding construction, that regexes could not match.

Center-embedding: tree

- Recall the center-embedding pattern that regexes and automata could not capture: NP $\underbrace{(\text{that NP})^k(\text{V})^k}_{\text{nested relative clauses}}$ VP, **for any** $k \geq 0$.

- We treat relative clauses (**RCs**) a bit like **complex adjectives**: they may attach to NPs, below the determiner.
- Trick: RCs themselves contain NPs, which themselves may contain other RCs! This makes RCs **recursive**: RCs may introduce new, nested RCs.
- Ultimately, this enables center-embedding.



Center-embedding: critical production rules

- Focusing on the nonterminal production rules:

$S \rightarrow$ DP VP

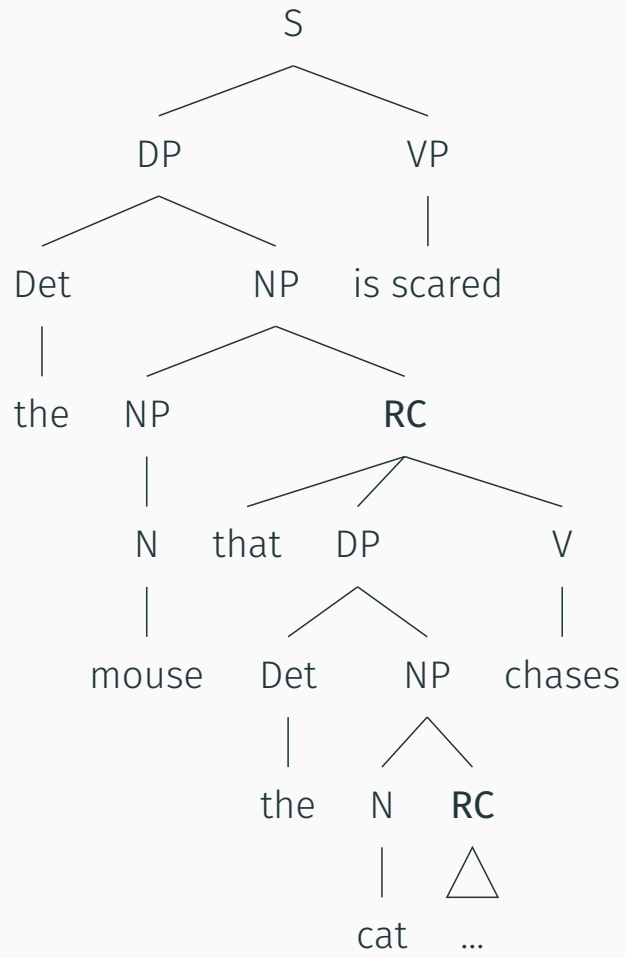
DP $\rightarrow$ Det NP

NP $\rightarrow$ NP **RC**

NP $\rightarrow$ **N**

RC $\rightarrow$ that **DP** V

- NPs introduce RCs or simply yield a N (crucial base case, for the derivation to stop at some point!)
- RCs in turn introduce DPs, which reintroduce NPs, which may reintroduce RCs!



Challenge: cross-serial dependencies

(12) Swiss German:

…mer em Hans es huus hälfed aastriiche.
…we Hans.DAT the house.ACC help paint.

'… we helped Hans paint the house.'

- In Swiss German, **predicates and their argument can be all jumbled up**!Nevertheless, predicates overtly **case-mark** their arguments, with Dative (in the case of *help*) and Accusative (in the case of *paint*).
- If we wanted to design a context-free grammar keeping track of predicate-argument dependencies (and of the resulting case-marking patterns), we'd struggle in the case of Swiss German, precisely **because the dependencies cross**.
- This pattern is quite rare cross-linguistically; dependencies tend to be nested. But it suggests that natural languages are in fact **more expressive than context-free grammars**.

Lab

https://colab.research.google.com/drive/11-deqVXa70AelTRruq7ZHZ3VXGflzkK_?usp=sharing